

ME1003-1 – ELEMENTS OF MECHANICAL ENGINEERING

Max. Marks: 100

ME1003

Part – B: Descriptive Answer type Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. _____ is not a conventional source of energy.
A) Coal
B) Petrol
C) Flowing water
D) Solar
2. _____ is a fire tube boiler.
A) Lancashire boiler
B) Babcock and Wilcox boiler
C) Stirling boiler
D) Yarrow boiler
3. The amount of heat required to raise the temperature of 1kg of water from 0°C to the saturation temperature T_s °C at a given constant pressure is defined as
A) Enthalpy of liquid
B) Latent heat
C) Enthalpy of evaporation
D) None of these
4. The boiling point of water _____ with the increase of pressure.
A) Decreases
B) Increases
C) Remains same
D) None of these
5. Prime mover in which the heat energy of the steam is transformed into mechanical energy.
A) Steam turbine
B) IC engine
C) Gas turbine
D) Water turbine
6. The output of the nozzle is
A) Low pressure and high velocity steam
B) Low pressure and low velocity steam
C) high pressure and high velocity steam
D) Low pressure and high velocity steam
7. The difference between saturation temperature and superheated temperature
A) Enthalpy of liquid
B) Latent heat
C) Enthalpy of evaporation
D) Degree of superheat
8. Indicated power of a 4-stroke engine is equal to
A) $4pLAN$
B) $pLAN/2$
C) $pLAN$
D) $2pLAN$
9. Which of the following does not relate to spark ignition engine?
A) Spark plug
B) Carburetor
C) Fuel injector
D) Ignition coil
10. The drive used for two shafts which are to be rotated in opposite direction is
A) Open belt drive
B) Cross belt drive
C) Stepped pulley
D) Chain drive
11. Chain drives with respect to belt drives give
A) More slip
B) Less slip
C) Equal slip
D) No slip
12. Heat is absorbed by refrigerant during a refrigeration cycle in a
A) Compressor
B) Condenser
C) Evaporator
D) Expansion valve
13. One tonne of refrigeration is
A) 210 kJ/min
B) 210 kJ/hour
C) 210 kJ/s
D) 210 MJ/min

14. In the arc welding process, the temperature of the arc can go up to a temperature of
 A) 1000 °C B) 3000 °C
 C) 6000 °C D) 9000 °C
15. Work – piece is held on
 A) Chuck B) Tail Stock
 C) Carriage D) Head Stock
16. Lathe center is used for
 A) Cutting B) Supporting
 C) Holding D) All of these
17. Automation is defined as the technology involved with the use of
 A) Mechanical system B) Electronics system
 C) Computer based system D) All the mentioned system
18. Which of these are not benefits of automation?
 A) Improved operational efficiency B) Decreases the productivity of the system
 C) Repetitive tasks can be completed faster D) Improved quality and consistency
19. The integration of mechanical system with the computer is referred as
 A) CAD system B) CAM System
 C) CAD/CAM System D) Electro mechanical system
20. The most obvious anthropomorphic characteristic of an industrial robot is
 A) Arm of the robot B) Drives used in robot
 C) Sensors used in robot D) Controller of the robot

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the process formation of steam with suitable sketches and graphs.	8	2	1	1
	b) Illustrate the working of Impulse steam turbine (De-Laval turbine) with the help of necessary sketches.	8	2	2	1
2.	a) Find the specific volume and enthalpy of 1 kg of steam at 10 Bar when (i) the dryness fraction is 0.6 and (ii) the steam is superheated to a temperature of 350°C. The specific heat of superheated steam is 2.25 kJ/kg.K. Given: $T_s = 179.9^\circ\text{C}$, $V_f = 0.001127 \text{ m}^3/\text{kg}$, $v_g = 0.194 \text{ m}^3/\text{kg}$, $h_f = 762.6 \text{ kJ/kg}$, $h_{fg} = 2013.6 \text{ kJ/kg}$, $h_g = 2776.2 \text{ kJ/kg}$.	8	3	1	2
	b) Describe the working of Francis reaction water turbine with a suitable figure.	8	2	2	1
3.	a) Classify various sources of energy. Briefly explain these sources of energy.	8	1	1	1
	b) Explain the working of a centrifugal pump with help of a figure.	8	2	2	1
Unit – II					
4.	a) The following observations were made during a trial on a 2-stroke petrol engine. Cylinder diameter = 20 cm; Stroke of the piston = 30 cm; Crankshaft speed = 400 rpm; Brake load = 80 kg; Brake drum diameter = 1.6 m; Mean effective pressure = 8 bar; Petrol oil consumption = 0.15 m ³ /min; Specific gravity of petrol = 0.8; Calorific value of petrol = 42,000 kJ/kg; Find: (i) Brake power, (ii) Indicated power, (iii) Mechanical efficiency, (iv) Indicated thermal efficiency.	8	3	3	2
	b) Explain the method of gas welding with necessary sketches. Also briefly mention steps followed in soldering.	8	2	4	1

5. a)	Describe with a sketch the working of a 2-stroke diesel engine.	8	2	3	1
b)	The driven pulley of 500 mm diameter of a belt drive runs at 350 rpm. The angle of lap is 160° and the coefficient of friction between the belt material and the pulley is 0.3. Find the power transmitted if the initial tension is not to exceed 15 kN.	8	2	4	2
6. a)	Describe working of Vapor compression refrigeration system with necessary sketches.	8	2	3	1
b)	Explain with figures the open & crossed belt drives and simple & compound gear trains.	8	1	4	1

Unit – III

7. a)	Write short notes on the following:				
i)	Open and closed loop mechatronic systems,				
ii)	Fixed automation,				
iii)	Programmable automation,				
iv)	Flexible automation.	8	1	5	1
b)	Explain the following Lathe operations:				
i)	Facing,				
ii)	Taper Turning,				
iii)	Thread Cutting,				
iv)	Plain turning.	8	1	5	1
8. a)	Explain the following machining operations with sketches:				
i)	End Milling,				
ii)	Drilling,				
iii)	Surface grinding,				
iv)	Tapping.	8	1	5	1
b)	Define Robotics. Describe the classifications of robots. Also list applications of robots.	8	1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
